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Properties of Liquids

In this chapter we discuss the various units of measurement employed in liquid pipeline hydraulics and proceed to cover the more important properties of liquids that affect hydraulic calculations. The importance of specific gravity, viscosity of pure liquids and mixtures will be analyzed and the concepts will be illustrated with sample problems. This chapter forms the foundation for all calculations involving pipeline pressure drops and horsepower requirements in subsequent chapters. In [Appendix A](#) you will find tables listing properties of commonly used liquids such as water and petroleum products.

2.1 Units of Measurement

Before we discuss liquid properties it would be appropriate to identify the different units of measurement used in pipeline hydraulics calculations. Over the years the English-speaking world adopted so-called English units of measurement, while most other European and Asian countries adopted the metric system of units.

The English system of units (referred to in the United States as Customary U.S. units) derives from the old foot-pound-second (FPS) and foot-slug-second (FSS) systems that originated in England. The basic units

are foot for length, slug for mass, and second for measurement of time. In the past, the FPS system used pound for mass. Since force, a derived unit, was also measured in pounds, there was evidently some confusion. To clarify, the terms pound-mass (lbm) and pound-force (lbf) were introduced. Numerically, the weight (which is a force due to gravity) of 1 pound-mass was equal to 1 pound-force. However, the introduction of slug for the unit of mass resulted in the adoption of pound exclusively for the unit of force. Thus, in the FSS system which is now used in the United States, the unit of mass is slug. The relationship between a slug, lbf, and lbm will be explained later in this chapter.

In the metric system, originally known as the centimeter-gram-second (CGS) system, the corresponding units for length, mass, and time were centimeter, gram, and second respectively. In later years, a modified metric system called the meter-kilogram-second (MKS) system emerged. In MKS units, the meter was used for the measurement of length and kilogram for the measurement of mass. The measurement for time remained the second for all systems of units.

The scientific and engineering communities have attempted during the last four decades to standardize on a universal system of units worldwide. Through the International Standards Organization (ISO), a policy for an International System of Units (SI) was formulated. The SI units are also known as *Système Internationale* units.

The conversion from the older system of units to SI units has advanced at different rates in different countries. Most countries of Western Europe and all of Eastern Europe, Russia, India, China, Japan, Australia, and South America have adopted SI units completely. In North America, Canada and Mexico have adopted SI units almost completely. However, engineers and scientists in these countries use both SI units and English units due to their business dealings with the United States. In the United States, SI units are used increasingly in colleges and the scientific community. However, the majority of work is still done using the English units referred to sometimes as Customary U.S. units.

The Metric Conversion Act of 1975 accelerated the adoption of the SI system of units in the United States. The American Society of Mechanical Engineers (ASME), the American Society of Civil Engineers (ASCE), and other professional societies and organizations have assisted in the process of conversion from English to SI units using the respective Institutions' publications. For example, ASME through the ASME Metric Study Committee published a series of articles in the *Mechanical Engineering* magazine to help engineers master the SI system of units.

In the United States, the complete changeover to SI units has not materialized fast enough. Therefore in this transition phase, engineering

students, practicing engineers, technicians, and scientists must be familiar with the different systems of units such as English, metric CGS, metric MKS, and SI. In this book we will use both English units (Customary U.S.) and the SI system of units.

Units of measurement are generally divided into three classes as follows:

- Base units
- Supplementary units
- Derived units

By definition, base units are dimensionally independent. These are units of length, mass, time, electric current, temperature, amount of substance, and luminous intensity.

Supplementary units are those used to measure plain angles and solid angles. Examples include the radian and steradian.

Derived units are those that are formed by combination of base units, supplementary units, and other derived units. Examples of derived units are those of force, pressure, and energy.

2.1.1 Base Units

In the English (Customary U.S.) system of units, the following base units are used:

Length	foot (ft)
Mass	slug (slug)
Time	second (s)
Electric current	ampere (A)
Temperature	degree Fahrenheit (°F)
Amount of substance	mole (mol)
Luminous intensity	candela (cd)

In SI units, the following base units are defined:

Length	meter (m)
Mass	kilogram (kg)
Time	second (s)
Electric current	ampere (A)
Temperature	kelvin (K)
Amount of substance	mole (mol)
Luminous intensity	candela (cd)

2.1.2 Supplementary Units

Supplementary units in both the English and SI systems of units are:

Plane angle	radian (rad)
Solid angle	steradian (sr)

The radian is defined as the plane angle between two radii of a circle with an arc length equal to the radius. Thus, it represents the angle of a sector of a circle with the arc length the same as its radius.

The steradian is the solid angle having its apex at the center of a sphere such that the area of the surface of the sphere that it cuts out is equal to that of a square with sides equal to the radius of this sphere.

2.1.3 Derived Units

Derived units are generated from a combination of base units, supplementary units, and other derived units. Examples of derived units include those of area, volume, etc.

In English units the following derived units are used:

Area	square inch (in. ²), square foot (ft ²)
Volume	cubic inch (in. ³), cubic foot (ft ³), gallon (gal), barrel (bbl)
Speed/velocity	foot per second (ft/s)
Acceleration	foot per second per second (ft/s ²)
Density	slug per cubic foot (slugs/ft ³)
Specific weight	pound per cubic foot (lb/ft ³)
Specific volume	cubic foot per pound (ft ³ /lb)
Dynamic viscosity	pound second per square foot (lb-s/ft ²)
Kinematic viscosity	square foot per second (ft ² /s)
Force	pound (lb)
Pressure	pound per square inch (lb/in. ² or psi)
Energy/work	foot pound (ft lb)
Quantity of heat	British thermal units (Btu)
Power	horsepower (HP)
Specific heat	Btu per pound per °F (Btu/lb/°F)
Thermal conductivity	Btu per hour per foot per °F (Btu/hr/ft/°F)

In SI units the following derived units are used:

Area	square meter (m ²)
Volume	cubic meter (m ³)
Speed/Velocity	meter/second (m/s)
Acceleration	meter per second per second (m/s ²)

Density	kilogram per cubic meter (kg/m^3)
Specific volume	cubic meter per kilogram (m^3/kg)
Dynamic viscosity	pascal second (Pa-s)
Kinematic viscosity	square meter per second (m^2/s)
Force	newton (N)
Pressure	newton per square meter or pascal (Pa)
Energy/work	newton meter or joule (J)
Quantity of heat	joule (J)
Power	joule per second or watt (W)
Specific heat	joule per kilogram per K (J/kg/K)
Thermal conductivity	joule/second/meter/kelvin (J/s/m/K)

Many other derived units are used in both English and SI units. A list of the more commonly used units in liquid pipeline hydraulics and their conversions is given in [Appendix A](#).

2.2 Mass, Volume, Density, and Specific Weight

Several properties of liquids that affect liquid pipeline hydraulics will be discussed here. In steady-state hydraulics of liquid pipelines, the following properties are important: mass, volume, density, and specific weight.

2.2.1 Mass

Mass is defined as the quantity of matter. It is independent of temperature and pressure. Mass is measured in slugs (slugs) in English units or kilograms (kg) in SI units. In the past mass was used synonymously with weight. Strictly speaking weight depends upon the acceleration due to gravity at a certain geographic location and therefore is considered to be a force. Numerically mass and weight are interchangeable in the older FPS system of units. For example, a mass of 10 lbm is equivalent to a weight of 10 lbf. To avoid this confusion, in English units the slug has been adopted as the unit of mass. One slug is equal to 32.17 lb. Therefore, if a drum contains 55 gal of crude oil and weighs 410 lb, the mass of oil will be the same at any temperature and pressure. Hence the statement “conservation of mass.”

2.2.2 Volume

Volume is defined as the space occupied by a given mass. In the case of the 55 gal drum above, 410 lb of crude oil occupies the volume of the drum. Therefore the crude oil volume is 55 gal. Consider a solid block of ice measuring 12 in. on each side. The volume of this block of ice is $12 \times 12 \times 12$ or 1728 cubic inches or 1 cubic foot (ft^3). The volume of a

certain petroleum product contained in a circular storage tank 100 ft in diameter and 50 ft high may be calculated as follows, assuming the liquid depth is 40 ft:

$$\text{Liquid volumes} = (\pi/4) \times 100 \times 100 \times 40 = 314,160 \text{ ft}^3$$

Liquids are practically incompressible, take the shape of their container and have a free surface. The volume of a liquid varies with temperature and pressure. However, since liquids are practically incompressible, pressure has negligible effect on volume. Thus, if the liquid volume measured at 50 pounds per square inch (psi) is 1000 gal, its volume at 1000 psi will not be appreciably different, provided the liquid temperature remained constant. Temperature, however, has a more significant effect on volume. For example, the 55 gal volume of liquid in a drum at a temperature of 60°F will increase to a slightly higher value (such as 56 gal) when the liquid temperature increases to 100°F. The increase in volume per unit temperature rise depends on the coefficient of expansion of the liquid. When measuring petroleum liquids, for the purpose of custody transfer, it is customary to correct volumes to a fixed temperature such as 60°F. Volume correction factors from American Petroleum Institute (API) publications are commonly used in the petroleum industry.

In the petroleum industry, it is customary to measure volume in gallons or barrels. One barrel is equal to 42 U.S. gallons. The Imperial gallon as used in the United Kingdom is a larger unit, approximately 20% larger than the U.S. gallon. In SI units volume is generally measured in cubic meters (m³) or liters (L).

In a pipeline transporting crude oil or refined petroleum products, it is customary to talk about the “line fill volume” of the pipeline. The volume of liquid contained between two valves in a pipeline can be calculated simply by knowing the internal diameter of the pipe and the length of pipe between the two valves. By extension the total volume, or the line fill volume, of the pipeline can be easily calculated.

As an example, if a 16 in. pipeline, 0.250 in. wall thickness, is 5000 ft long from one valve to another, the line fill for this section of pipeline is

$$\begin{aligned} \text{Line fill volume} &= (\pi/4) \times (16 - 2 \times 0.250)^2 \times 5000 \\ &= 943,461.75 \text{ ft}^3 \text{ or } 168,038 \text{ bbl} \end{aligned}$$

The above calculation is based on conversion factors of:

- 1728 in.³ per ft³
- 231 in.³ per gallon
- 42 gallons per barrel.

In [Chapter 6](#) we discuss a simple formula for determining the line fill volume of a pipeline.

The volume flow rate in a pipeline is generally expressed in terms of cubic feet per second (ft^3/s), gallons per minute (gal/min), barrels per hour (bbl/hr), and barrels per day (bbl/day) in English units. In SI units, volume flow rate is measured in cubic meters per hour (m^3/hr) and liters per second (L/s).

It must be noted that since the volume of a liquid varies with temperature, the inlet flow rate and the outlet volume flow rate may be different in a long-distance pipeline, even if there are no intermediate injections or deliveries. This is due to the fact that the inlet flow rate may be measured at an inlet temperature of 70°F to be $5000 \text{ bbl}/\text{hr}$ and the corresponding flow rate at the pipeline terminus, 100 miles away, may be measured at an outlet temperature different from the inlet temperature. The temperature difference is due to heat loss or gain between the pipeline liquid and the surrounding soil or ambient conditions. Generally, significant variation in temperature is observed when pumping crude oils or other products that are heated at the pipeline inlet. In refined petroleum products and other pipelines that are not heated, temperature variations along the pipeline are insignificant. In any case if the volume measured at the pipeline inlet is corrected to a standard temperature such as 60°F , the corresponding outlet volume can also be corrected to the same standard temperature. With temperature correction it can be assumed that the same flow rate exists throughout the pipeline from inlet to outlet, provided of course there are no intermediate injections or deliveries along the pipeline.

By the principle of conservation of mass, the mass flow rate at the inlet will equal that at the pipeline outlet since the mass of liquid does not change with temperature or pressure.

2.2.3 Density

Density of a liquid is defined as the mass per unit volume. Customary units for density are slugs/ ft^3 in English units and kg/m^3 in SI units. This is also referred to as mass density. The weight density is defined as the weight per unit volume. This term is more commonly called specific weight and will be discussed next.

Since mass does not change with temperature or pressure, but volume varies with temperature, we can conclude that density will vary with temperature. Density and volume are inversely related since density is defined as mass per unit volume. Therefore, with increase in temperature liquid volume increases while its density decreases. Similarly, with reduction in temperature, liquid volume decreases and its density increases.

2.2.4 Specific Weight

Specific weight of a liquid is defined as the weight per unit volume. It is measured in lb/ft^3 in English units and N/m^3 in SI units.

If a 55 gal drum of crude oil weighs 410 lb (excluding the weight of the drum), the specific weight of crude oil is

$$(410/55) \text{ or } 7.45 \text{ lb/gal}$$

Similarly, consider the 5000 ft pipeline discussed in Section 2.2.2. The volume contained between the two valves was calculated to be 168,038 bbl. If we use the specific weight calculated above, we can estimate the weight of liquid contained in the pipeline as

$$7.45 \times 42 \times 168,038 = 52,579,090 \text{ lb or } 26,290 \text{ tons}$$

Similar to density, specific weight varies with temperature. Therefore, with increase in temperature specific weight will decrease. With reduction in temperature, liquid volume decreases and its specific weight increases.

The customary unit for specific weight is lb/ft^3 and lb/gal in the English units. The corresponding SI unit of specific weight is N/m^3 .

For example, water has a specific weight of 62.4 lb/ft^3 or 8.34 lb/gal at 60°F . A typical gasoline has a specific weight of 46.2 lb/ft^3 or 6.17 lb/gal at 60°F .

Although density and specific weight are dimensionally different, it is common to use the term density instead of specific weight and vice versa when calculating hydraulics of liquid pipelines. Thus you will find that the density of water and specific weight of water are both expressed as 62.4 lb/ft^3 .

2.3 Specific Gravity and API Gravity

Specific gravity of a liquid is the ratio of its density to the density of water at the same temperature and therefore has no units (dimensionless). It is a measure of how heavy a liquid is compared with water. By definition, the specific gravity of water is 1.00.

The term relative density is also used to compare the density of a liquid with another liquid such as water. It should be noted that both densities must be measured at the same temperature for the comparison to be meaningful.

At 60°F , a typical crude oil has a density of 7.45 lb/gal compared with a water density of 8.34 lb/gal . Therefore, the specific gravity of crude oil at 60°F is

$$\text{Specific gravity} = 7.45/8.34 \text{ or } 0.8933$$

Specific gravity, like density, varies with temperature. As temperature increases, both density and specific gravity decrease. Similarly, a decrease in temperature causes the density and specific gravity to increase in value. As with volume, pressure has very little effect on liquid specific gravity as long as pressures are within the range of most pipeline applications.

In the petroleum industry, it is customary to use units of °API for gravity. The API gravity is a laboratory-determined scale that compares the density of the liquid with that of water at 60°F (fixed at a value of 10). All liquids lighter than water will have API values higher than 10. Thus gasoline has an API gravity of 60 while a typical crude oil may be 35°API.

The API gravity versus the specific gravity relationship is as follows:

$$\text{Specific gravity (Sg)} = 141.5 / (131.5 + \text{API}) \quad (2.1)$$

or

$$\text{API} = 141.5 / \text{Sg} - 131.5 \quad (2.2)$$

Substituting an API value of 10 for water in Equation (2.1), yields as expected the specific gravity of 1.00 for water. It is seen from the above equation that the specific gravity of the liquid cannot be greater than 1.076 in order to result in a positive value of API.

Another scale of gravity for liquids heavier than water is known as the Baume scale. This scale is similar to the API scale with the exception of 140 and 130 being used in place of 141.5 and 131.5 respectively in Equations (2.1) and (2.2).

As another example, assume the specific gravity of gasoline at 60°F is 0.736. Therefore, the API gravity of gasoline can be calculated from Equation (2.2) as follows:

$$\text{API gravity} = 141.5 / 0.736 - 131.5 = 60.76^\circ \text{API}$$

If diesel fuel is reported to have an API gravity of 35, the specific gravity can be calculated from Equation (2.1) as follows:

$$\text{Specific gravity} = 141.5 / (131.5 + 35) = 0.8498$$

It must be noted that API gravity is always referred to at 60°F. Therefore in the Equations (2.1) and (2.2), specific gravity must also be measured at 60°F. Hence, it is meaningless to say that the API of a liquid is 35°API at 70°F.

API gravity is measured in the laboratory by the method described in ASTM D1298, using a properly calibrated glass hydrometer. The reader is referred to the *API Manual of Petroleum Measurements* for further discussion on API gravity.

2.3.1 Specific Gravity Variation with Temperature

It was mentioned earlier that the specific gravity of a liquid varies with temperature. It increases with a decrease in temperature and vice versa.

For the commonly encountered range of temperatures in liquid pipelines, the specific gravity of a liquid varies linearly with temperature. In other words, the specific gravity versus temperature can be expressed in the form of the following equation:

$$S_T = S_{60} - a(T - 60) \quad (2.3)$$

where

S_T = Specific gravity at temperature T

S_{60} = Specific gravity at 60°F

T = Temperature, °F

a = A constant that depends on the liquid

In Equation (2.3) the specific gravity S_T at temperature T is related to the specific gravity at 60°F by a straight-line relationship. Since the terms S_{60} and a are unknown quantities, two sets of specific gravities at two different temperatures are needed to determine the specific gravity versus temperature relationship. If the specific gravity at 60°F and the specific gravity at 70°F are known we can substitute these values in Equation (2.3) to obtain the unknown constant a . Once the value of a is known, we can easily calculate the specific gravity of the liquid at any other temperature using Equation (2.3). An example will illustrate how this is done.

Some handbooks such as the *Hydraulic Institute Engineering Data Book* and the *Flow of Fluids through Valves, Fittings and Pipes* (see [References](#)) provide specific gravity versus temperature curves from which the specific gravity of most liquids can be calculated at any temperature.

Example Problem 2.1

The specific gravity of gasoline at 60°F is 0.736. The specific gravity at 70°F is 0.729. What is the specific gravity at 50°F?

Solution

Using Equation (2.3), we can write

$$0.729 = 0.736 - a(70 - 60)$$

Solving for a , we get

$$a = 0.0007$$

We can now calculate the specific gravity at 50°F using Equation (2.3) as

$$S_{50}=0.736-0.0007(50-60)=0.743$$

2.3.2 Specific Gravity of Blended Liquids

Suppose a crude oil of specific gravity 0.895 at 70°F is blended with a lighter crude oil of specific gravity 0.815 at 70°F, in equal volumes. What will be the specific gravity of the blended mixture? Common sense suggests that since equal volumes are used, the resultant mixture should have a specific gravity that is the average of the two liquids, or

$$(0.895+0.815)/2=0.855$$

This is indeed the case, since specific gravity of a liquid is simply related to the mass and the volume of each liquid.

When two or more liquids are mixed homogeneously, the specific gravity of the resultant liquid can be calculated using the weighted average method. Thus, 10% of liquid A with a specific gravity of 0.85 when blended with 90% of liquid B that has a specific gravity of 0.89 results in a blended liquid with a specific gravity of

$$(0.1 \times 0.85) + (0.9 \times 0.89) = 0.886$$

It must be noted that when performing the above calculations, both specific gravities must be measured at the same temperature.

Using the above approach, the specific gravity of a mixture of two or more liquids can be calculated from the following equation:

$$S_b = \frac{(Q_1 \times S_1) + (Q_2 \times S_2) + (Q_3 \times S_3) + \dots}{Q_1 + Q_2 + Q_3 + \dots} \quad (2.4)$$

where

S_b = Specific gravity of the blended liquid

Q_1, Q_2, Q_3 , etc. = Volume of each component

S_1, S_2, S_3 , etc. = Specific gravity of each component

The above method of calculating the specific gravity of a mixture of two or more liquids cannot be directly applied when the gravities are expressed in °API values. If the component gravities of a mixture are given in °API we must first convert API values to specific gravities before applying Equation (2.4).

Example Problem 2.2

Three liquids A, B, and C are blended together in the ratio of 15%, 20%, and 65% respectively. Calculate the specific gravity of the blended liquid if the individual liquids have the following specific gravities at 70°F:

Specific gravity of liquid A: 0.815

Specific gravity of liquid B: 0.850

Specific gravity of liquid C: 0.895

Solution

Using Equation (2.4) we can calculate the specific gravity of the blended liquid as

$$S_b = (15 \times 0.815 + 20 \times 0.850 + 65 \times 0.895) / 100 = 0.874$$

2.4 Viscosity

Viscosity is a measure of sliding friction between successive layers of a liquid that flows in a pipeline. Imagine several layers of liquid that constitute a flow between two fixed parallel horizontal plates. A thin layer adjacent to the bottom plate will be at rest or zero velocity. Each subsequent layer above this will have a different velocity compared with the layer below. This variation in the velocity of the liquid layers results in a velocity gradient. If the velocity is V at the layer that is located a distance of y from the bottom plate, the velocity gradient is approximately:

$$\text{Velocity gradient} = V/y \quad (2.5)$$

If the variation of velocity with distance is not linear, using calculus we can write more accurately that

$$\text{Velocity gradient} = \frac{dV}{dy} \quad (2.6)$$

where dV/dy represents the rate of change of velocity with distance or the velocity gradient.

Newton's law states that the shear stress between adjacent layers of a flowing liquid is proportional to the velocity gradient. The constant of proportionality is known as the absolute (or dynamic) viscosity of the liquid.

$$\text{Shear stress} = (\text{Viscosity}) (\text{Velocity gradient})$$

The absolute viscosity of a liquid is measured in lb-s/ft² in English units and pascal-s in SI units. Other commonly used units of absolute viscosity are the poise and centipoise (cP).

The kinematic viscosity is defined as the absolute viscosity of a liquid divided by its density at the same temperature.

$$\nu = \mu / \rho \quad (2.7)$$

where

ν =Kinematic viscosity

μ =Absolute viscosity

ρ =Density

The units of kinematic viscosity are ft²/s in English units and m²/s in SI units. See [Appendix A](#) for conversion of units. Other commonly used units for kinematic viscosity are the stoke and centistoke (cSt). In the petroleum industry, two other units for kinematic viscosity are also used. These are saybolt seconds universal (SSU) and saybolt seconds furol (SSF). When expressed in these units, kinematic viscosity represents the time taken for a fixed volume of a liquid to flow through an orifice of defined size. Both absolute and kinematic viscosities vary with temperature. As temperature increases, liquid viscosity decreases and vice versa. However, unlike specific gravity, viscosity versus temperature is not a linear relationship. We will discuss this in Section 2.4.1.

Viscosity also varies somewhat with pressure. Significant variation in viscosity is found when pressures are several thousand psi. In most pipeline applications, viscosity of a liquid does not change appreciably with pressure.

For example, the viscosities of Alaskan North Slope (ANS) crude oil may be reported as 200 SSU at 60°F and 175 SSU at 70°F. Viscosity in SSU and SSF may be converted to their equivalent in centistokes using the following equations:

Conversion from SSU to centistokes

$$\text{Centistokes} = 0.226(\text{SSU}) - 195/(\text{SSU}) \text{ for } 32 \leq \text{SSU} \leq 100 \quad (2.8)$$

$$\text{Centistokes} = 0.220(\text{SSU}) - 135/(\text{SSU}) \text{ for } \text{SSU} > 100 \quad (2.9)$$

Conversion from SSF to centistokes

$$\text{Centistokes} = 2.24(\text{SSF}) - 184/(\text{SSF}) \text{ for } 25 < \text{SSF} \leq 40 \quad (2.10)$$

$$\text{Centistokes} = 2.16(\text{SSF}) - 60/(\text{SSF}) \text{ for } \text{SSF} > 40 \quad (2.11)$$

Example Problem 2.3

Use the above equations to convert the viscosity of ANS crude oil from 200 SSU to its equivalent in centistokes.

Solution

Using Equation (2.9):

$$\begin{aligned}\text{Centistokes} &= 0.220 \times 200 - 135/200 \\ &= 43.33 \text{ cSt}\end{aligned}$$

The reverse process of converting from viscosity in cSt to its equivalent in SSU using Equations (2.8) and (2.9) is not quite so direct. Since Equations (2.8) and (2.9) are valid for certain ranges of SSU values, we need first to determine which of the two equations to use. This is difficult since the SSU value is unknown. Therefore we will have to assume that the SSU value to be calculated falls in one of the two ranges and proceed to calculate by trial and error. We will have to solve a quadratic equation to determine the SSU value for a given viscosity in cSt. An example will illustrate this method.

Example Problem 2.4

Suppose we are given a liquid viscosity of 15 cSt. Calculate the corresponding viscosity in SSU.

Solution

Let us assume that the calculated value in SSU is approximately $5 \times 15 = 75$ SSU. This is a good approximation, since the SSU value is generally about 5 times the corresponding viscosity value in cSt. Since the assumed SSU value is 75, we need to use Equation (2.8) for converting between cSt and SSU.

Substituting 15 cSt in Equation (2.8) gives

$$15 = 0.226(\text{SSU}) - 195/(\text{SSU})$$

Replacing SSU with variable x , the above Equation becomes, after transposition

$$15x = 0.226x^2 - 195$$

Rearranging we get

$$0.226x^2 - 15x - 195 = 0$$

Solving for x, we get

$$x = [15 + (15 \times 15 + 4 \times 0.226 \times 195)^{1/2}] / (2 \times 0.226) = 77.5$$

Therefore, the viscosity is 77.5 SSU.

2.4.1 Viscosity Variation with Temperature

The viscosity of a liquid decreases as the liquid temperature increases and vice versa. For gases, the viscosity increases with temperature. Thus, if the viscosity of a liquid at 60°F is 35 cSt, as the temperature increases to 100°F, the viscosity could drop to a value of 15 cSt. The variation of liquid viscosity with temperature is not linear, unlike the variation of specific gravity with temperature discussed in Section 2.3.1. The variation of viscosity with temperature is found to be logarithmic in nature.

Mathematically, we can state the following:

$$\text{Log}_e(v) = A - B(T) \quad (2.12)$$

where

v = Viscosity of liquid, cSt

T = Absolute temperature, °R or °K

$$T = (t + 460)^\circ\text{R} \text{ if } t \text{ is in } ^\circ\text{F} \quad (2.13)$$

$$T = (t + 273)^\circ\text{K} \text{ if } t \text{ is in } ^\circ\text{C} \quad (2.14)$$

A and B are constants that depend on the specific liquid.

It can be seen from Equation (2.12) that a graphic plot of $\log_e(v)$ against the temperature T will result in a straight line with a slope of $-B$. Therefore, if we have two sets of viscosity versus temperature values for a liquid we can determine the values of A and B by substituting the viscosity and temperature values in Equation (2.12). Once A and B are known we can calculate the viscosity at any other temperature using Equation (2.12). An example will illustrate this.

Example Problem 2.5

Suppose we are given the viscosities of a liquid at 60°F and 100°F as 43 cSt and 10 cSt. What is the viscosity at 80°F?

Solution

We will use Equation (2.12) to calculate the values of A and B first:

$$\text{Log}_e(43) = A - B(60 + 460)$$

and

$$\text{Log}_e(10)=A-B(100+460)$$

Solving the above two equations for A and B results in

$$A=22.7232 \quad B=0.0365$$

Having found A and B, we can now calculate the viscosity of this liquid at any other temperature using Equation (2.12). To calculate the viscosity at 80°F:

$$\text{Log}_e(v)=22.7232-0.0365(80+460)=3.0132$$

$$\text{Viscosity at } 80^\circ\text{F}=20.35 \text{ cSt}$$

In addition to Equation (2.12), several researchers have put forth various equations that attempt to correlate the viscosity variation of petroleum liquids with temperature. The most popular and accurate of the formulas is the one known as the ASTM method. In this method, also known as the ASTM D341 chart method, special graph paper with logarithmic scales is used to plot the viscosity of a liquid at two known temperatures. Once the two points are plotted on the chart and a line drawn connecting them, the viscosity at any intermediate temperature can be interpolated. To some extent, values of viscosity may also be extrapolated from this chart. This is shown in [Figure 2.1](#).

In the following paragraphs we discuss how to calculate the viscosity variations with temperature, using the ASTM method, without using the special logarithmic graph paper.

$$\text{Log Log}(Z)=A-B \text{ Log}(T) \quad (2.15)$$

where

Log=logarithm to base 10

Z depends on the viscosity of the liquid v

v =viscosity of liquid, cSt

T=Absolute temperature, °R or K

A and B are constants that depend on the specific liquid.

The variable Z is defined as follows:

$$Z=(v+0.7+C-D) \quad (2.16)$$

where C and D are

$$C=\exp[-1.14883-2.65868(v)] \quad (2.17)$$

$$D=\exp[-0.0038138-12.5645(v)] \quad (2.18)$$

C, D, and Z are all functions of the kinematic viscosity v .

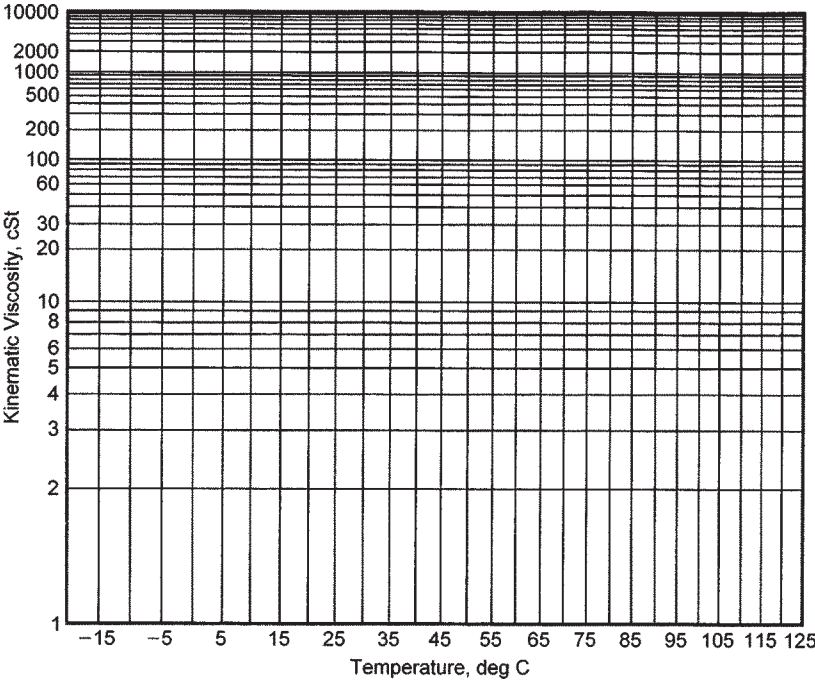


Figure 2.1 ASTM D341—viscosity temperature chart.

Given two sets of temperature viscosity values (T_1, ν_1) and (T_2, ν_2) we can calculate the corresponding values of C, D, and Z from Equations (2.17), (2.18), and (2.16).

We can then come up with two equations using the pairs of (T_1, Z_1) and (T_2, Z_2) values by substituting these values into Equation (2.15) as shown below:

$$\text{Log Log } (Z_1)=A-B \text{ Log}(T_1) \tag{2.19}$$

$$\text{Log Log } (Z_2)=A-B \text{ Log}(T_2) \tag{2.20}$$

From the above equations, the two unknown constants A and B can be easily calculated, since T_1, Z_1 and T_2, Z_2 values are known.

The following example will illustrate this approach for viscosity temperature variation.

Example Problem 2.6

A certain liquid has a temperature versus viscosity relationship as given below:

Temperature, °F	60	180
Viscosity, cSt	750	25

- Calculate the constants A and B that define the viscosity versus temperature correlation for this liquid using Equation (2.15).
- What is the estimated viscosity of this liquid at 85°F?

Solution

(a) At the first temperature 60°F, C, D, and Z are calculated using Equations (2.17), (2.18), and (2.16):

$$C_1 = \exp[-1.14883 - 2.65868 \times 750] = 0$$

$$D_1 = \exp[-0.0038138 - 12.5645 \times 750] = 0$$

$$Z_1 = (750 + 0.7) = 750.7$$

Similarly at the second temperature of 180°F, the corresponding values of C, D, and Z are calculated to be:

$$C_2 = \exp[-1.14883 - 2.65868 \times 25] = 0$$

$$D_2 = \exp[-0.0038138 - 12.5645 \times 25] = 0$$

$$Z_2 = (25 + 0.7) = 25.7$$

Substituting in Equation (2.19) we get

$$\text{Log Log}(750.7) = A - B \text{ Log}(60 + 460)$$

or

$$0.4587 = A - 2.716B$$

$$\text{Log Log}(25.7) = A - B \text{ Log}(180 + 460)$$

or

$$0.1492 = A - 2.8062B$$

Solving for A and B, we get

$$A = 9.778$$

$$B = 3.4313$$

(b) At a temperature of 85°F, using Equation (2.15) we get

$$\text{Log Log}(Z)=A-B \text{ Log}(85+460)$$

$$\text{Log Log}(Z)=9.778-3.4313 \times 2.7364=0.3886$$

$$Z=279.78$$

Therefore,

$$\text{Viscosity at } 85^{\circ}\text{F}=279.78-0.7=279.08 \text{ cSt}$$

2.4.2 Viscosity of Blended Products

Suppose a crude oil of viscosity 10 cSt at 60°F is blended with a lighter crude oil of viscosity 30 cSt at 60°F, in equal volumes. What will be the viscosity of the blended mixture? We cannot average the viscosities as we did with the specific gravities of mixtures earlier. This is due to the nonlinear nature of viscosity with mass and volumes of liquids.

When blending two or more liquids, the specific gravity of the blended product can be calculated directly, by using the weighted average approach as demonstrated in Section 2.3.2. However, the viscosity of a blend of two or more liquids cannot be calculated by simply using the ratio of each component. Thus if 20% of liquid A of viscosity 10 cSt is blended with 80% of liquid B with a viscosity of 30 cSt, the blended viscosity is *not* the following

$$0.2 \times 10 + 0.8 \times 30 = 26 \text{ cSt}$$

In fact, the actual blended viscosity would be 23.99 cSt as will be demonstrated in the following section.

The viscosity of a blend of two or more products can be estimated using the following equation:

$$\sqrt{V_b} = \frac{Q_1 + Q_2 + Q_3 + \dots}{(Q_1/\sqrt{V_1}) + (Q_2/\sqrt{V_2}) + (Q_3/\sqrt{V_3})} \quad (2.21)$$

where

V_b = Viscosity of blend, SSU

Q_1, Q_2, Q_3 , etc. = Volumes of each component

V_1, V_2, V_3 , etc. = Viscosity of each component, SSU

Since Equation (2.19) requires the component viscosities to be in SSU, we cannot use this equation to calculate the blended viscosity when viscosity is less than 32 SSU (1.0 cSt).

Another method of calculating the viscosity of blended products has been in use in the pipeline industry for over four decades. This method is

referred to as the Blending Index method. In this method a Blending Index is calculated for each liquid based on its viscosity. Next the Blending Index of the mixture is calculated from the individual blending indices by using the weighted average of the composition of the mixture. Finally, the viscosity of the blended mixture is calculated using the Blending Index of the mixture. The equations used are given below:

$$H=40.073-46.414 \log_{10} (V+B) \quad (2.22)$$

$$B=0.931(1.72)^V \text{ for } 0.2 < V < 1.5 \quad (2.23)$$

$$B=0.6 \text{ for } V \geq 1.5 \quad (2.24)$$

$$H_m=[H_1(\text{pct1})+H_2(\text{pct2})+H_3(\text{pct3})+\dots]/100 \quad (2.25)$$

where

$H, H_1, H_2, \dots$ = Blending Index of liquids

H_m = Blending Index of mixture

B = Constant in Blending Index equation

V = Viscosity, cSt

pct1, pct2, etc. = Percentage of liquids 1, 2, etc. in blended mixture.

Example Problem 2.7

Calculate the blended viscosity obtained by mixing 20% of liquid A with a viscosity of 10 cSt and 80% of liquid B with a viscosity of 30 cSt at 70°F.

Solution

First, convert the given viscosities to SSU to use Equation (2.21). The viscosity of liquid A is calculated using Equations (2.8) and (2.9):

$$10 = 0.226(V_A) - \frac{195}{V_A}$$

Rearranging we get

$$0.226 V_A^2 - 10 V_A - 195 = 0$$

Solving the quadratic equation for V_A we get

$$V_A = 58.90 \text{ SSU}$$

Similarly, viscosity of liquid B is

$$V_B = 140.72 \text{ SSU}$$

From Equation (2.21), the blended viscosity is

$$\begin{aligned}\sqrt{V_{\text{blend}}} &= \frac{20 + 80}{(20/\sqrt{58.9}) + (80/\sqrt{140.72})} \\ &= 10.6953\end{aligned}$$

Therefore the viscosity of the blend is

$$V_{\text{blend}} = 114.39 \text{ SSU}$$

or

$$\begin{aligned}\text{Viscosity of blend} &= 23.99 \text{ cSt} \quad \text{after converting from} \\ &\quad \text{SSU to cSt using Equation} \quad (2.9)\end{aligned}$$

A graphical method is also available to calculate the blended viscosities of two petroleum products using ASTM D341. This method involves using a logarithmic chart with viscosity scales on the left and right side of the paper. The horizontal axis is for selecting the percentage of each product as shown in [Figure 2.2](#). This chart is also available in handbooks such as *Flow of Fluids through Valves, Fittings and Pipes* and the *Hydraulic Institute Engineering Data Book* (see [References](#)). It must be noted that the viscosities of both products must be plotted at the same temperature.

Using this method, the blended viscosity of two products at a time is calculated and the process repeated for multiple products. Thus if three products are blended in the ratios of 10%, 20%, and 70%, we would first calculate the viscosity of the blend using the first two liquids, considering 10 parts of liquid A mixed with 20 parts of liquid B. This means that the blend would be calculated on the basis of one-third of liquid A and two-thirds of liquid B. Next this blended liquid will be mixed with liquid C in the proportions of 30% and 70% respectively.

2.5 Vapor Pressure

Vapor pressure of a liquid is defined as the pressure at a given temperature at which the liquid and vapor exist in equilibrium. The normal boiling point of a liquid can thus be defined as the temperature at which the vapor pressure equals the atmospheric pressure. In the laboratory the vapor pressure is measured at a fixed temperature of 100°F and is then reported as the Reid vapor pressure. The vapor pressure of a liquid increases with temperature. Charts are available to determine the actual vapor pressure of a liquid at any temperature once its Reid vapor pressure is known. The

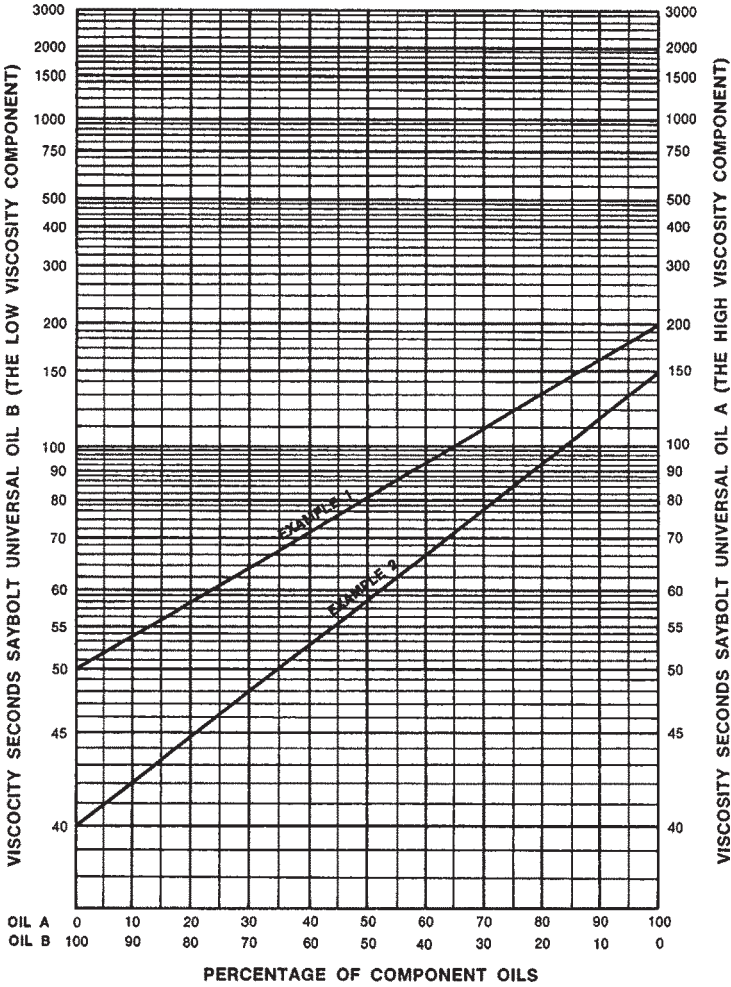


Figure 2.2 Viscosity blending chart.

reader is referred to *Fluid Flow through Valves, Fittings and Pipes* for vapor pressure charts.

The importance of vapor pressure will be evident when we discuss the operation of centrifugal pumps on pipelines. To prevent cavitation of pumps, the liquid vapor pressure at the flowing temperature must be taken into

account in the calculation of net positive suction head (NPSH) available at the pump suction. Centrifugal pumps are discussed in [Chapter 7](#).

2.6 Bulk Modulus

The bulk modulus of a liquid is a measure of its compressibility. It is defined as the pressure required to produce a unit change in its volume. Mathematically, bulk modulus is expressed as

$$\text{Bulk modulus } K = V dP/dV \quad (2.26)$$

where dV is the change in volume corresponding to a change in pressure of dP .

The units of bulk modulus, K , are psi or kPa. For most liquids the bulk modulus is approximately in the range of 250,000 to 300,000 psi. The fairly high number demonstrates the incompressibility of liquids.

Let us demonstrate the incompressibility of liquids by performing a calculation using bulk modulus. Assume that the bulk modulus of a petroleum product is 250,000 psi. To calculate the pressure required to change the volume of a given quantity of liquid by 1% we would proceed as follows. From Equation (2.26), with some rearrangement,

$$\text{Bulk modulus} = \text{change in pressure} / (\text{change in volume} / \text{volume})$$

Therefore

$$250,000 = \text{change in pressure} / (0.01)$$

Therefore

$$\text{Change in pressure} = 2500 \text{ psi}$$

It can be seen from the above that a fairly large pressure is required to produce a very small (1%) change in the liquid volume. Hence we say that liquids are fairly incompressible.

Bulk modulus is used in line pack calculations and transient flow analysis. There are two bulk modulus values used in practice: isothermal and adiabatic. The bulk modulus of a liquid depends on temperature, pressure, and specific gravity. The following empirical equations, also known as ARCO formulas, may be used to calculate the bulk modulus.

2.6.1 Adiabatic Bulk Modulus

$$K_a = A + B(P) - C(T)^{1/2} - D(API) - E(API)^2 + F(T)(API) \quad (2.27)$$

where

$$A=1.286 \times 10^6$$

$$B=13.55$$

$$C=4.122 \times 10^4$$

$$D=4.53 \times 10^3$$

$$E=10.59$$

$$F=3.228$$

P=Pressure, psig

T=Temperature, °R

API=API gravity of liquid

2.6.2 Isothermal Bulk Modulus

$$K_i = A + B(P) - C(T)^{1/2} + D(T)^{3/2} - E(API)^{3/2} \quad (2.28)$$

where

$$A=2.619 \times 10^6$$

$$B=9.203$$

$$C=1.417 \times 10^5$$

$$D=73.05$$

$$E=341.0$$

P=Pressure, psig

T=Temperature, °R

API=API gravity of liquid

For a typical crude oil of 35°API gravity at 1000 psig pressure and 80°F temperature, the bulk modulus values calculated from Equations (2.27) and (2.28) are:

Adiabatic bulk modulus=231,426 psi

Isothermal bulk modulus=181,616 psi

The bulk modulus of water at 70°F is 320,000 psi.

2.7 Fundamental Concepts of Fluid Flow

In this section we discuss fundamental concepts of fluid flow that will set the stage for the succeeding chapter. The basic principles of continuity and energy equations are introduced first.

2.7.1 Continuity

One of the fundamental concepts that must be satisfied in any type of pipe flow is the principle of continuity of flow. This principle states that the

total amount of fluid passing through any section of a pipe is fixed. This may also be thought of as the principle of conservation of mass. Basically, it means that liquid is neither created nor destroyed as it flows through a pipeline. Since mass is the product of the volume and density, we can write the following equation for continuity:

$$M = \text{Vol} \times \rho = \text{Constant} \quad (2.29)$$

where

M = Mass flow rate at any point in the pipeline, slugs/s

Vol = Volume flow rate at any point in the pipeline, ft^3/s

ρ = Density of liquid at any point in the pipeline, slugs/ ft^3

Since the volume flow rate at any point in a pipeline is the product of the area of cross-section of the pipe and the average liquid velocity, we can rewrite Equation (2.29) as follows:

$$M = A \times V \times \rho = \text{Constant} \quad (2.30)$$

where

M = Mass flow rate at any point in the pipeline, slugs/s

A = Area of cross-section of pipe, ft^2

V = Average liquid velocity, ft/s

ρ = Density of liquid at any point in the pipeline, slugs/ ft^3

Since liquids are generally considered to be incompressible and therefore density does not change appreciably, the continuity equation reduces to

$$AV = \text{Constant} \quad (2.31)$$

2.7.2 Energy Equation

The basic principle of conservation of energy applied to liquid hydraulics is embodied in Bernoulli's equation, which simply states that the total energy of the fluid contained in the pipeline at any point is a constant. Obviously, this is an extension of the principle of conservation of energy which states that energy is neither created nor destroyed, but transformed from one form to another.

Consider the pipeline shown in [Figure 2.3](#) that depicts flow from point A to point B with the elevation of point A being Z_A and elevation at B being Z_B above some chosen datum. The pressure in the liquid at point A is P_A and that at B is P_B . Assuming a general case, where the pipe diameter at A may be different from that at B, we will designate the velocities at A and B to be V_A and V_B respectively. Consider a particle of the liquid of weight

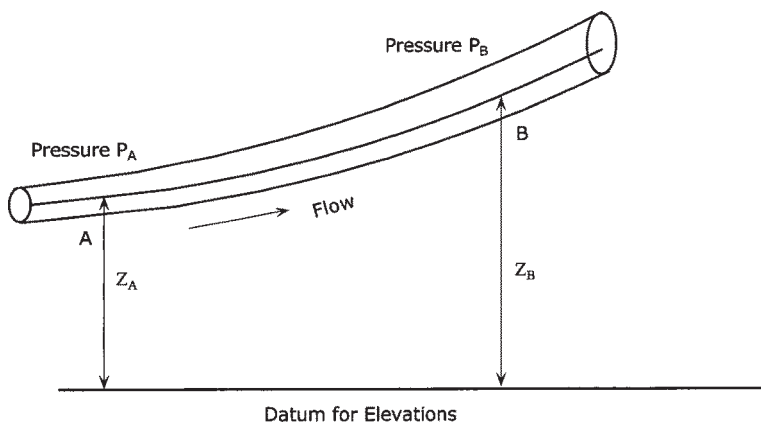


Figure 2.3 Energy of a liquid in pipe flow.

W at point A in the pipeline. This liquid particle at A may be considered to possess a total energy E that consists of three components:

Energy due to position, or potential energy = WZ_A

Energy due to pressure, or pressure energy = WP_A/γ

Energy due to velocity, or kinetic energy = $W(V_A^2)/2g$

where

γ = Specific weight of the liquid

We can thus state that

$$E = WZ_A + WP_A/\gamma + WV_A^2/2g \quad (2.32)$$

Dividing by W throughout, we get the total energy per unit weight of liquid as

$$H_A = Z_A + P_A/\gamma + V_A^2/2g \quad (2.33)$$

where

H_A = total energy per unit weight at point A

Considering the same liquid particle as it arrives at point B, the total energy per unit weight at B is

$$H_B = Z_B + P_B/\gamma + V_B^2/2g \quad (2.34)$$

Due to conservation of energy

$$H_A = H_B$$

Therefore,

$$Z_A + P_A/\gamma + V_A^2/2g = Z_B + P_B/\gamma + V_B^2/2g \quad (2.35)$$

Equation (2.35) is one form of Bernoulli's equation for fluid flow.

In real-world pipeline transportation, there is energy loss between point A and point B, due to friction in the pipe. We include the energy loss due to friction by modifying Equation (2.35) as follows:

$$Z_A + P_A/\gamma + V_A^2/2g = Z_B + P_B/\gamma + V_B^2/2g + \Sigma h_L \quad (2.36)$$

where

Σh_L = all the head losses between points A and B, due to friction. In Bernoulli's equation (2.35), we must also include any energy added to the liquid, such as when there is a pump between points A and B. Thus the left-hand side of the equation will have a positive term added to it that will represent the energy generated by a pump. Equation (2.36) will be modified as follows to include a pump at point A that will add a certain amount of pump head to the liquid:

$$Z_A + P_A/\gamma + V_A^2/2g + H_P = Z_B + P_B/\gamma + V_B^2/2g + \Sigma h_L \quad (2.37)$$

where

H_P = pump head added to the liquid at point A

In [Chapter 3](#), we further explore the concepts of pressure, velocity, flow rates, and energy lost due to pipe friction.

2.8 Summary

In this chapter we discussed the more important properties of liquids that determine the nature of liquid flow in pipelines. The specific gravity and viscosity of liquids were explained along with how to calculate these properties in liquid mixtures and at various temperatures. We also introduced the basic concepts of liquid flow consisting of the continuity equation and the energy equation embodied in Bernoulli's equation.

2.9 Problems

- 2.9.1 Calculate the specific weight and specific gravity of a liquid that weighs 312 lb, contained in a volume of 5.9 ft³. Assume water weighs 62.4 lb/ft³.
- 2.9.2 The specific gravities of a liquid at 60°F and 100°F are reported to be 0.895 and 0.815 respectively. Determine the specific

- gravity of the liquid at 85°F. Assume a linear relationship between specific gravity and temperature.
- 2.9.3 The gravity of a petroleum product is 59°API. Calculate the corresponding specific gravity at 60°F.
- 2.9.4 The viscosity of a liquid at 70°F is 45 cSt. Express this viscosity in SSU. If the specific gravity at 70°F is 0.885, determine the absolute or dynamic viscosity.
- 2.9.5 The viscosities of a crude oil at 60°F and 100°F are 40 cSt and 15 cSt respectively. Using the ASTM correlation method, calculate the viscosity of this product at 80°F.
- 2.9.6 Two liquids are blended to form a homogeneous mixture. Liquid A has a specific gravity of 0.815 at 70°F and a viscosity of 15 cSt at 70°F. At the same temperature, liquid B has a specific gravity of 0.85 and viscosity of 25 cSt. If 20% of liquid A is blended with 80% of liquid B, calculate the specific gravity and viscosity of the blended product.
- 2.9.7 Using the viscosity blending chart, calculate the blended viscosity for two liquids as follows:

Product	Percentage	Viscosity (SSU)
Liquid A	15	50
Liquid B	85	200

- 2.9.8 If liquid A with a viscosity of 40 SSU is blended with liquid B of viscosity 150 SSU, what percentage of each component would be required to obtain a blended viscosity of 46 SSU?
- 2.9.9 In [Figure 2.3](#), consider the pipe to be 20 in. diameter. The liquid is water with a specific gravity of 1.00. The point A is at elevation 100 ft and B is at elevation 200 ft. The pressure at A is 500 psi and that at B is 400 psi. Specific weight of water is 62.34 lb/ft³. Write down the Bernoulli equation for energy conservation between points A and B.